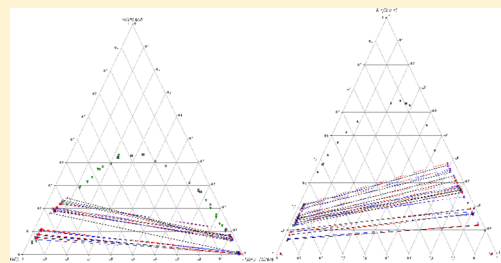


Evaluation of the Performance of Four Solvents for the Liquid–Liquid Extraction of Acrylic Acid from Water

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ABSTRACT: In this work, experimental liquid–liquid equilibrium data for four ternary systems containing {water + acrylic acid + solvent} were determined at 298.15 K and atmospheric pressure. The experimental data were also compared with the values correlated by the nonrandom two-liquid (NRTL) and universal quasichemical (UNIQUAC) models. Good quantitative agreement was obtained with these models. In general, the average deviation from the NRTL model is slightly greater than those from the UNIQUAC model. The reliability of the experimental tie-line data was confirmed by using the Othmer–Tobias correlation.



1. INTRODUCTION

Acrylic acid is a commonly used, yet extremely valuable, chemical intermediate used to produce polymers, textiles, and numerous other industrial and consumer products.¹ Typically, acrylic acid is prepared by the oxidation of propylene. In this case, acrylic acid is produced as a diluted aqueous solution, which must be separated and purified. Modern plants are based on the gas phase catalytic oxidation of propylene via acrolein. The reaction can be carried out in single- or two-step processes, but the latter is favored because of higher yields. The oxidation of propylene produces acrolein, acrylic acid, acetaldehyde, and carbon oxides. Acrylic acid from the primary oxidation can be recovered while the acrolein is fed to a second step to make acrylic acid. Purification can be carried out by azeotropic distillation. Because of the thermal stability of the acrylic acid and the important energy consumed for its purification using the distillation technique, liquid–liquid extraction seems to be a potential alternative.

In the literature, several solvents can be found to extract acrylic acid from aqueous solution through liquid–liquid equilibria (LLE) data. However, (liquid + liquid) equilibrium data in aqueous systems with acrylic acid are extremely scarce. For instance, Linek et al.² and Chubarov et al.^{3,4} studied this kind of systems with several organic acids and esters. Batistell et al.⁵ studied extraction of acrylic acid with 1-butanol, 2-butanol, or 1-pentanol were determined at 293.2 K, 303.2 K, and 313.2 K, and Pinto et al.⁶ measured experimental data for liquid–liquid ternary system {water + acrylic acid + 4-methyl-2-pentanone} at 298.15 K and 1 atm.

The distribution coefficient and selectivity are the most important parameters that govern solvent selection. The desired properties of solvents are a high distribution coefficient, good selectivity toward solute, and little or no miscibility with feed solution. Also, the solvent should be easily recoverable for recycling.

In this work, the performance of four solvents is evaluated for the extraction of acid acrylic from water. Experimental liquid–liquid equilibrium data for four ternary systems containing {water + acrylic acid + (4-methyl-2-pentanone, toluene, *o*-xylene, or *p*-xylene)} were determined at 298.15 K and at atmospheric pressure. Experimental data were correlated using the non-random two-liquid (NRTL) and universal quasichemical (UNIQUAC) activity coefficient models.

2. EXPERIMENTAL SECTION

Chemicals. The physical properties of the compounds used are shown in Table 1. Acrylic acid with a purity > 0.99, 4-methyl-2-pentanone (MIBK), and toluene (Aldrich > 0.995), *o*-xylene (Aldrich > 0.995), and *p*-xylene (Aldrich > 0.995), respectively, were used. Water was purified by a Milli-Q system with a resistivity less than 18 Ω ·cm.

Table 1. Molar Mass (*M*), Density (ρ), and Purity of Compounds Used in This Work

compound	<i>M</i> /(g·mol ^{−1})	ρ /(kg·m ^{−3})	
		this work	lit.
water	18.02	997.02	998.21 ⁷
acrylic acid	72.06	1059.17	N.A. ^a
MIBK	100.16	798.09	796.43 ⁸
toluene	92.14	862.34	862.34 ⁹
<i>o</i> -xylene	106.165	875.52	875.52 ¹⁰
<i>p</i> -xylene	106.165	856.65	856.66 ¹⁰

^aN.A.: not available.

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Table 2. Equation for Density (ρ) as a Function of Water and Solvent Mass Fractions (w), Standard Deviations of Fit (σ), and Correlation of Coefficients (R^2)

equation	$\sigma/(\text{kg}\cdot\text{m}^{-3})$	R^2
water (1) + acrylic acid (2) + MIBK (3) $\rho(T = 298.15 \text{ K})/(\text{kg}\cdot\text{m}^{-3}) = 1059.69 - 40.69w_1 - 267.23w_3$	0.337	0.9931
water (1) + acrylic acid (2) + toluene (3) $\rho(T = 298.15 \text{ K})/(\text{kg}\cdot\text{m}^{-3}) = 1059.69 - 18.99w_1 - 209.60w_3$	0.613	0.9914
water (1) + acrylic acid (2) + <i>o</i> -xylene (3) $\rho(T = 298.15 \text{ K})/(\text{kg}\cdot\text{m}^{-3}) = 1059.69 + 21.36w_1 - 192.21w_3$	0.366	0.9911
water (1) + acrylic acid (2) + <i>p</i> -xylene (3) $\rho(T = 298.15 \text{ K})/(\text{kg}\cdot\text{m}^{-3}) = 1059.69 - 25.44w_1 - 228.01w_3$	0.730	0.9882

Table 3. Cloud Points (Mass Fraction w) and Density (ρ) for {Water (1) + Acrylic Acid (2) + Solvent (3)} at $T = 298.15 \text{ K}$

w_1	w_2	w_3	$\rho/(\text{kg}\cdot\text{m}^{-3})$	w_1	w_2	w_3	$\rho/(\text{kg}\cdot\text{m}^{-3})$
Water (1) + Acrylic Acid (2) + MIBK (3)				Water (1) + Acrylic Acid (2) + <i>o</i> -Xylene (3)			
0.915	0.069	0.016	1003.877				
0.910	0.070	0.020	1002.178	0.704	0.281	0.014	1042.151
0.745	0.217	0.038	1017.122	0.474	0.504	0.022	1057.934
0.674	0.272	0.054	1017.745	0.424	0.542	0.035	1058.858
0.660	0.278	0.062	1019.782	0.365	0.581	0.053	1062.641
0.603	0.311	0.086	1014.057	0.300	0.646	0.054	1061.855
0.459	0.385	0.156	1000.676	0.254	0.675	0.071	1057.796
0.401	0.406	0.194	993.278	0.224	0.684	0.092	1053.994
0.330	0.435	0.235	987.560	0.188	0.682	0.130	1044.222
0.281	0.430	0.289	973.664	0.094	0.631	0.275	1007.475
0.223	0.423	0.354	955.597	0.073	0.579	0.347	992.907
0.174	0.382	0.444	934.445	0.056	0.547	0.396	983.107
0.039	0.198	0.763	857.004	0.049	0.532	0.418	978.966
0.038	0.199	0.763	848.639	0.034	0.494	0.472	968.762
0.017	0.053	0.930	810.479	0.022	0.459	0.519	962.454
Water (1) + Acrylic Acid (2) + Toluene (3)				0.017	0.413	0.570	950.420
0.896	0.097	0.006	1019.122	0.008	0.299	0.692	913.954
0.802	0.190	0.008	1028.273	0.007	0.144	0.848	907.956
				Water (1) + Acrylic Acid (2) + <i>p</i> -Xylene (3)			
0.763	0.214	0.023	1035.054	0.844	0.146	0.010	1015.651
0.702	0.270	0.028	1040.413	0.785	0.200	0.015	1022.182
0.501	0.436	0.063	1055.055	0.766	0.216	0.018	1023.055
0.410	0.543	0.048	1048.867	0.566	0.409	0.026	1042.691
0.291	0.629	0.080	1051.332	0.395	0.579	0.026	1060.027
0.270	0.637	0.093	1047.139	0.347	0.612	0.040	1057.796
0.265	0.638	0.097	1043.938	0.347	0.616	0.037	1051.224
0.237	0.647	0.116	1041.346	0.301	0.659	0.040	1050.203
0.202	0.636	0.162	1029.029	0.255	0.679	0.067	1052.619
0.087	0.559	0.353	982.970	0.163	0.685	0.151	1031.364
0.061	0.513	0.426	964.850	0.091	0.630	0.278	990.528
0.041	0.472	0.488	952.492	0.024	0.402	0.575	930.026
0.025	0.418	0.557	940.435	0.014	0.378	0.608	919.700
0.023	0.380	0.598	935.357	0.013	0.362	0.625	916.773
0.019	0.378	0.603	931.286				
0.014	0.347	0.638	927.332				

Procedure. Ternary liquid–liquid equilibrium data were determined by stirring weighted water–solvent mixtures in an equilibrium cell equipped with magnetic stirring and a jacket for circulation of the isothermal fluid ($298.15 \pm 0.05 \text{ K}$). The third component was added from a microburet until transition from turbidity to a clear solution was observed. In this way, the central portions of the binodal curves were obtained. Taking into account that the accuracy with which the transition point can be observed is better as the angle between the tangent to the binodal curve in the transition point and the line passing through the

composition of the binary system is nearer to a right angle. Water–acrylic acid and acrylic acid–solvent mixtures were titrated with the remaining component to obtain the end portions of the binodal curves.

The quantities of the third component were determined by weighing the mixture after the transition point was reached.

The uncertainty in the observed values was determined by the accuracy of the observation of the transition point, since the quantities of titrating component and titrated mixtures were accurately determined by weighing. The uncertainty through the

Table 4. (Liquid + Liquid) Equilibrium Experimental Data (Mass Fraction w), Density (ρ), Acrylic Distribution Coefficient (β), and Solvent Selectivity (S) for {Water (1) + Acrylic Acid (2) + Organic Solvent (3)} at Atmospheric Pressure and 298.15 K

aqueous phase				organic phase				β	S
w_1	w_2	w_3	$\rho(\text{kg}\cdot\text{m}^{-3})$	w_1	w_2	w_3	$\rho(\text{kg}\cdot\text{m}^{-3})$		
Water (1) + Acrylic Acid (2) + MIBK (3)									
0.990	0.002	0.008	997.307	0.020	0.075	0.905	817.780	37.500	1856.250
0.987	0.004	0.009	998.048	0.025	0.085	0.890	820.629	21.250	838.950
0.980	0.008	0.012	998.453	0.025	0.105	0.870	825.437	13.125	514.500
0.970	0.008	0.022	1000.152	0.035	0.180	0.785	846.247	22.500	623.571
0.920	0.060	0.020	1001.881	0.045	0.195	0.760	851.885	3.250	66.444
0.914	0.061	0.025	1004.490	0.080	0.245	0.675	873.872	4.016	45.887
0.911	0.077	0.012	1001.447	0.035	0.200	0.765	852.757	2.597	67.607
0.910	0.070	0.020	1003.205	0.065	0.230	0.705	866.383	3.286	46.000
0.890	0.080	0.030	1007.306	0.070	0.235	0.695	867.806	2.938	37.348
Water (1) + Acrylic Acid (2) + Toluene (3)									
0.970	0.000	0.030	1005.182	0.020	0.000	0.980	868.776		
0.860	0.120	0.020	1022.064	0.020	0.060	0.920	878.559	0.500	21.500
0.810	0.170	0.020	1028.195	0.020	0.080	0.900	883.291	0.471	19.059
0.790	0.190	0.020	1031.207	0.020	0.100	0.880	886.155	0.526	20.789
0.710	0.260	0.030	1038.607	0.015	0.120	0.865	891.688	0.462	21.846
0.705	0.270	0.025	1039.441	0.020	0.130	0.850	891.376	0.481	16.972
0.700	0.270	0.030	1039.500	0.020	0.140	0.840	892.731	0.519	18.148
0.690	0.280	0.030	1041.699	0.020	0.150	0.830	894.163	0.536	18.482
0.660	0.310	0.030	1044.831	0.033	0.150	0.817	896.775	0.484	9.677
0.640	0.330	0.030	1045.538	0.020	0.160	0.820	897.350	0.485	15.515
0.620	0.345	0.035	1049.231	0.015	0.190	0.795	900.517	0.551	22.763
0.590	0.375	0.035	1050.910	0.020	0.205	0.775	904.007	0.547	16.127
Water (1) + Acrylic Acid (2) + <i>o</i> -Xylene (3)									
0.985	0.000	0.015	1000.087	0.020	0.000	0.980	881.275		
0.900	0.075	0.025	1013.043	0.005	0.145	0.850	883.160	1.933	348.000
0.885	0.100	0.015	1015.455	0.010	0.190	0.800	892.758	1.900	168.150
0.865	0.120	0.015	1018.092	0.010	0.205	0.785	895.284	1.708	147.771
0.825	0.160	0.015	1025.301	0.010	0.235	0.755	899.225	1.469	121.172
0.820	0.165	0.015	1026.066	0.010	0.240	0.750	899.716	1.455	119.273
0.790	0.190	0.020	1030.363	0.020	0.250	0.730	906.520	1.316	51.974
0.770	0.210	0.020	1033.287	0.010	0.265	0.725	907.729	1.262	97.167
0.765	0.215	0.020	1034.298	0.015	0.265	0.720	908.486	1.233	62.860
0.730	0.250	0.020	1039.098	0.010	0.275	0.715	909.932	1.100	80.300
Water (1) + Acrylic Acid (2) + <i>p</i> -Xylene (3)									
0.985	0.000	0.015	996.688	0.020	0.000	0.980	867.520		
0.900	0.080	0.020	1009.044	0.025	0.150	0.825	877.994	1.875	67.500
0.850	0.130	0.020	1015.141	0.025	0.175	0.800	881.694	1.346	45.769
0.845	0.135	0.020	1016.141	0.025	0.185	0.790	883.827	1.370	46.319
0.830	0.150	0.020	1017.475	0.025	0.205	0.770	886.486	1.367	45.373
0.680	0.300	0.020	1032.257	0.020	0.310	0.670	906.620	1.033	35.133
0.780	0.200	0.020	1022.594	0.020	0.200	0.780	885.453	1.000	39.000

whole curve was less than 0.05 weight fraction acrylic acid for all systems.

After equilibrium was reached, the densities of the ternary mixtures were measured with a vibrating tube densimeter Anton Paar DMA 60 at 298.15 ± 0.1 K with an accuracy of ± 0.01 $\text{kg}\cdot\text{m}^{-3}$, and with these values a calibrating curve density versus mass fraction of acrylic acid was drawn.

For the determination of tie-line data, different weighted ternary mixtures with compositions within the heterogeneous region were shaken vigorously in the equilibrium cell for at least 2 h at constant temperature and left for about 1 h, allowing phase separation and equilibrium establishment. The upper solvent rich layer and the lower water rich layer were sampled with different syringes.

Samples were analyzed by measuring their densities, and the compositions were obtained with the aid of the calibrating curve. In this way, the two end points of a tie line were obtained. An independent material-balance checking was provided by the fact that the overall composition of an equilibrium mixture must lie on the straight line joining the equilibrium layers.

3. RESULTS AND DISCUSSION

The cloud-point density values, presented in Tables 3 and 4, were used to find expressions for density as a function of water (w_1) and solvent (w_3) mass fractions. These expressions are used to determine the unknown acrylic acid composition. The expressions found for density as a function of water and acrylic acid mass fractions for all systems studied here are present in

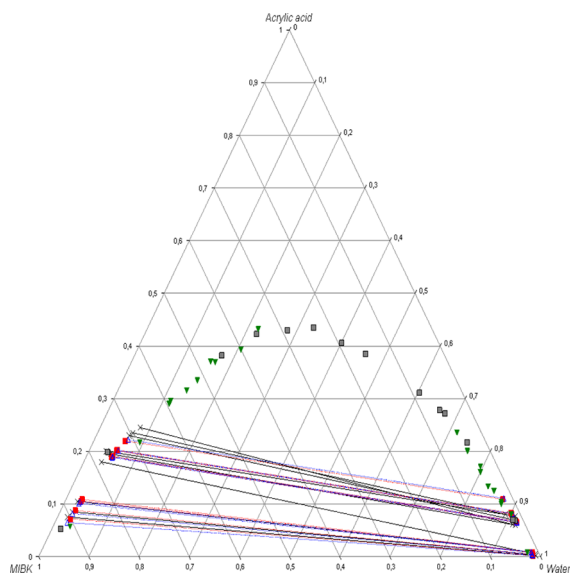


Figure 1. Tie lines for the ternary mixture {water + acrylic acid + MIBK} at 298.15 K. Experimental data: + and —, NRTL correlation; red ■ and red ---, UNIQUAC correlation; blue △ and blue ---, cloud points; green ▼, Pinto et al.⁶

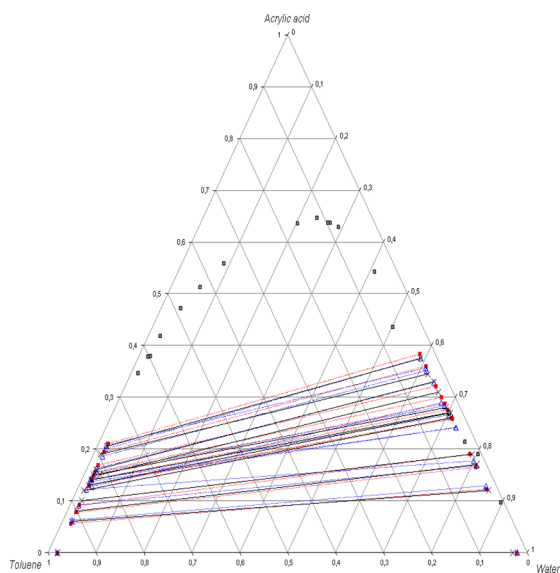


Figure 2. Tie lines for the ternary mixture {water + acrylic acid + toluene} at 298.15 K. Experimental data: + and —, NRTL correlation; red ■ and red ---, UNIQUAC correlation; blue △ and blue ---, cloud points.

Table 2. Also, in this table the fit standard deviation (σ) and the correlation coefficient (R^2) are shown for each equation.

Tables 3 and 4 list the cloud points and the compositions of the experimental tie-lines for the six investigated ternary systems at 298.15 K. The phase diagrams of these systems are shown in Figures 1 to 4. Values of solute distribution ratio β and selectivity S are also reported in Table 5. These parameters were calculated from experimental data to evaluate solvent extraction capacity according to equations:

$$\beta = \frac{x_1^{\text{solvent}}}{x_1^{\text{water}}} \quad (1)$$

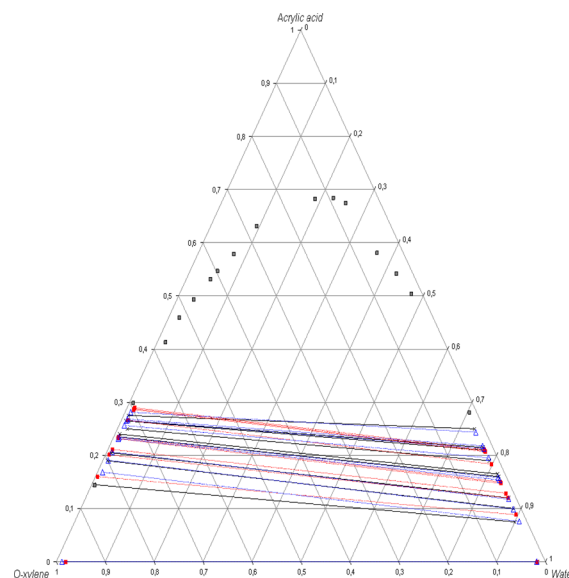


Figure 3. Tie lines for the ternary mixture {water + acrylic acid + *o*-xylene} at 298.15 K. Experimental data: + and —, NRTL correlation; red ■ and red ---, UNIQUAC correlation; blue △ and blue ---, cloud points.

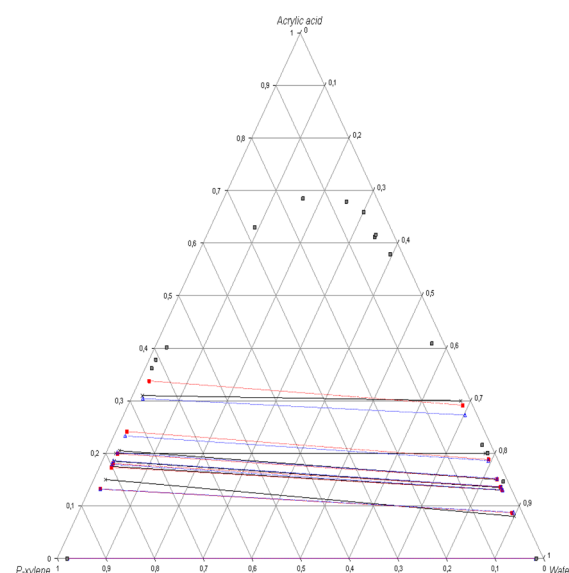


Figure 4. Tie lines for the ternary mixture {water + acrylic acid + *p*-xylene} at 298.15 K. Experimental data: + and —, NRTL correlation; red ■ and red ---, UNIQUAC correlation; blue △ and blue ---, cloud points.

Table 5. van der Waals Volume and Surface Parameters

	water	acrylic acid	MIBK	toluene	<i>o</i> -xylene	<i>p</i> -xylene
q_i	1.4000	2.4000	3.9520	2.9680	3.5360	3.5360
r_i	0.9200	2.6467	4.5959	3.9229	4.6579	4.6579

$$S = \frac{x_1^{\text{solvent}} \cdot x_2^{\text{water}}}{x_1^{\text{water}} \cdot x_2^{\text{solvent}}} \quad (2)$$

where x is the mole fraction, subscripts 1 and 2 refer to solute (acrylic acid, solvent) and to solvent (MIBK, toluene, *p*-xylene, or *o*-xylene), respectively, and the superscripts solvent and water indicate the solvent-rich phase and the water-rich phase,

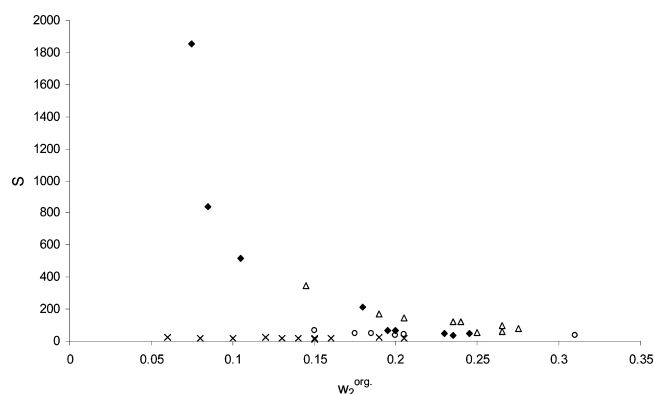


Figure 5. Selectivity S as a function of mole fraction of acrylic acid in the organic-rich phase for the four systems studied. $\blacklozenge$, MIBK; $\times$, toluene; $\triangle$, *o*-xylene; $\circ$, *p*-xylene.

respectively. To have an adequate separation, values of solute distribution ratios have to be higher than the unit.

Experimental liquid–liquid equilibrium data for the ternary system (water + acrylic acid + 4-methyl-2-pentanone) at 298.15 K were carried out to validate our apparatus. Results obtained are in good agreement with those measured by Pinto et al.⁶ All liquid–liquid equilibrium data for the ternary system reported in this paper form type I phase diagrams and present a partial miscibility gap relatively small. The evolution of the selectivities for each system {acrylic acid + water} as a function of the mole fraction of the solute is plotted in Figure 5.

For three studied systems, the solvents show distribution ratio largely higher than 1, which indicates very high acrylic acid solubility in these solvents. The experimental liquid–liquid equilibrium data for the ternary system (water + acrylic acid + 4-methyl-2-pentanone) indicates that the acrylic acid/water selectivities and the solute distribution ratios are relatively high. This solvent is particularly efficient for the extraction of acrylic acid at low concentration. Results obtained with the aromatic compounds show that the structure of the molecule plays an important role on their extractive performance. The alkyl chains grafted on the aromatic ring influence the interaction between the acrylic acid and the aromatic solvent. The performance of the three aromatics decreases according to: *o*-xylene > *p*-xylene > toluene. Results obtained indicate that both

Table 7. Binary Interaction Parameters for the Correlation with the UNIQUAC Model

	a_{ij}	a_{ji}	b_{ij}	b_{ji}
Water (1) + Acrylic Acid (2) + MIBK (3)				
1–2	33.785	7.326	−5841.294	−2371.904
1–3	0.000	0.000	−77.852	−468.350
2–3	−20.503	26.372	6539.399	−3999.004
Water (1) + Acrylic Acid (2) + Toluene (3)				
1–2	0.000	0.000	−159.150	−4848.208
1–3	0.000	0.000	−67.051	−385.486
2–3	−1.081	−1.414	−199.081	181.232
Water (1) + Acrylic Acid (2) + <i>o</i> -Xylene (3)				
1–2	0.255	−3.977	76.159	−1185.799
1–3	0.158	−1.702	47.238	−507.415
2–3	−0.568	0.093	−169.458	27.704
Water (1) + Acrylic Acid (2) + <i>p</i> -Xylene (3)				
1–2	0.000	0.000	1525.779	−4593.997
1–3	0.000	0.000	39.612	−154.675
2–3	−0.208	−0.828	−4584.300	1692.255

xylene may be used for the extraction of acrylic acid. While toluene has a good selectivity, the solvent presents very low distribution ratios.

The LLE data of the investigated ternary systems were correlated using the nonrandom two-liquid equation (NRTL) proposed by Renon and Prausnitz¹¹ and the UNiVersal QUAsi-Chemical (UNIQUAC) theory developed by Abrams and Prausnitz.¹²

NRTL Model. For the NRTL model, the activity coefficient γ_i for any component i of the ternary system, is given by:

$$\ln \gamma_i = \frac{\sum_{j=1}^m \tau_{ji} G_{ji} x_j}{G_{ii} x_i} + \sum_{j=1}^m \frac{x_j G_{ji}}{\sum_{l=1}^m G_{li} x_l} \left(\tau_{ij} \frac{\sum_{r=1}^m x_r \tau_{rj} G_{rj}}{\sum_{l=1}^m G_{lj} x_l} \right) \quad (3)$$

with $G_{ji} = \exp(-\alpha_{ji} \tau_{ji})$, $\tau_{ji} = a_{ji} + (b_{ji})/T$ and $\alpha_{ji} = \alpha_{ij} = \alpha$, where G is an energy parameter characterizing the interaction of species i and j . x_i is the mole fraction of component i . α is the nonrandomness parameter. Although α can be treated as an adjustable parameter, in this study α was set equal to 0.2 or 0.3 according to the literature.¹¹

Table 6. Binary Interaction Parameters for the Correlation with the NRTL Model

	a_{ij}	a_{ji}	b_{ij}	b_{ji}	α
Water (1) + Acrylic Acid (2) + MIBK (3)					
1–2	−1.86	4.147	−554.8317	1236.3425	0.3
1–3	1.2587	282.1266	−761.676	−12671.4629	0.2
2–3	−1.4942	−1.8912	−445.4881	−563.8511	0.3
Water (1) + Acrylic Acid (2) + Toluene (3)					
1–2	0	0	−1014.4276	1155.9795	0.3
1–3	627.0528	−247.8792	−28036.478	13718.666	0.2
2–3	0.6689	−1.8929	199.426	−564.3526	0.3
Water (1) + Acrylic Acid (2) + <i>o</i> -Xylene (3)					
1–2	0	0	−1398.7872	244.9172	0.3
1–3	−1.8428	0.4458	2244.3977	5394.0216	0.2
2–3	−33.4629	−16.7692	9981.8998	3027.8392	0.3
Water (1) + Acrylic Acid (2) + <i>p</i> -Xylene (3)					
1–2	−3.70908487	3.21344733	2041.70541	−169.3308	0.3
1–3	162.4774	2.7734	−605.771113	221.741847	0.2
2–3	4.06792009	5.78042529	−549.846173	−546.54536	0.3

Table 8. Root-Mean-Square Error (RMSE)^a

		RMSE %				RMSE %	
		NRTL	UNIQUAC			NRTL	UNIQUAC
Water (1) + Acrylic Acid (2) + 4-Methyl-2-pentanone (3)				Water (1) + Acrylic Acid (2) + Toluene (3)			
water-rich layer	water	0.9798	0.8977	water-rich layer	water	0.7328	1.1953
	acrylic acid	1.3345	1.3404		acrylic acid	0.7764	1.2621
	4-methyl-2-pentanone	0.6041	0.7199		toluene	0.3877	0.3547
organic-rich layer	water	1.1394	1.0490	organic-rich layer	water	0.4305	0.2516
	acrylic acid	3.0021	3.1390		acrylic acid	0.5384	1.1912
	4-methyl-2-pentanone	3.6609	3.7078		toluene	0.7006	1.3013
Water (1) + Acrylic Acid (2) + <i>o</i> -Xylene (3)				Water (1) + Acrylic Acid (2) + <i>p</i> -Xylene (3)			
water-rich layer	water	1.9708	0.4135	water-rich layer	water	0.6539	1.0128
	acrylic acid	1.9875	0.4175		acrylic acid	0.6116	1.1922
	<i>o</i> -xylene	0.3319	0.3306		<i>p</i> -xylene	0.0598	0.2108
organic-rich layer	water	0.2586	0.4508	organic-rich layer	water	0.1281	0.2696
	acrylic acid	1.6152	0.8283		acrylic acid	2.0271	1.4540
	<i>o</i> -xylene	1.5316	1.0084		<i>p</i> -xylene	2.1366	1.5960

$$^a\text{RMSE} = ((1/N)\sum_i (w_i^{\text{exptl}} - w_i^{\text{pred}})^2)^{1/2}.$$

Table 9. Hand and Othmer–Tobias Equation Coefficients (a, b) and Linear Correlation Coefficient (R^2) for Systems Studied in This Work

solvent	Hand			Othmer–Tobias		
	a_H	b_H	R^2	a_{OT}	b_{OT}	R^2
MIBK	0.3652	−0.0265	0.9363	0.5571	−0.1546	0.9126
toluene	0.9311	−0.4012	0.9720	1.1628	−0.4729	0.9646
<i>o</i> -xylene	0.4574	−0.1856	0.9871	0.4762	−0.1724	0.9768
<i>p</i> -xylene	0.5561	−0.1913	0.9276	0.5393	0.1777	0.9199

UNIQUAC Model. For the UNIQUAC model, the activity coefficient γ_i for any component i of the ternary system is given by:

$$\ln \gamma_i = \ln \frac{\Phi_i}{x_i} + \frac{z}{2} q_i \ln \frac{\theta_i}{\Phi_i} + l_i - \frac{\Phi_i}{x_i} \sum_{j=1}^m x_j l_j - q_i \ln(\theta_j \tau_{ji}) + q_i - q_i \sum_{j=1}^m \frac{\theta_j \tau_{ji}}{\sum_{k=1}^m \theta_k \tau_{kj}} \quad (4)$$

where $\Phi_i = (r_i x_i) / (\sum_{j=1}^m r_j x_j)$, $\theta_i = (q_i x_i) / (\sum_{j=1}^m q_j x_j)$, $l_i = (z/2)(r_j - q_j) - (r_j - 1)$ and $\tau_{ji} = \exp(a_{ji} + (b_{ji}/T))$.

Here, the lattice coordination number z is assumed to be equal to 10, r_i and q_i are respectively a relative volume and surface area of the pure component i . Parameters r_i and q_i are respectively relative to molecular van der Waals volumes and molecular surface areas. They are calculated as the sum of the group volume and group area parameters R_k and Q_k :

$$r_i = \sum_k v_k^i R_k \quad \text{and} \quad q_i = \sum_k v_k^i Q_k \quad (5)$$

where v_k^i is the number of groups of type k in molecule i . The group parameters R_k and Q_k are obtained from van der Waals group volumes and surface areas, V_k and A_k taken from the universal functional activity coefficient (UNIFAC) group contributions:¹³

$$R_k = \frac{V_k}{15.17} \quad \text{and} \quad Q_k = \frac{A_k}{2.5 \cdot 10^9} \quad (6)$$

The values of 15.17 and $2.5 \cdot 10^9$ are respectively the standard segment volume and standard segment area of a methylene group.¹³

This means that a total of four adjustable parameters per binary, a_{ij} , a_{ji} , b_{ij} and b_{ji} have to be fitted for both models. The binary interaction parameters of {water + 4-methyl-2-pentanone} are taken from the literature.⁶

For the binary LLE problem, the maximum likelihood method and Britt–Luecke algorithm¹⁴ were used. The maximum likelihood objective function is:

$$F_{\text{obj}} = \sum_{k=1}^N \sum_{i=1}^3 \{ (x_{i,k}^{\text{I,exp}} - x_{i,k}^{\text{I,calc}})^2 + (x_{i,k}^{\text{II,exp}} - x_{i,k}^{\text{II,calc}})^2 \} \quad (7)$$

where N is the number of tie lines in the data set, $x_{i,k}^{\text{I,exp}}$ and $x_{i,k}^{\text{I,calc}}$ are the experimental and calculated mole fractions of one phase, and $x_{i,k}^{\text{II,exp}}$ and $x_{i,k}^{\text{II,calc}}$ are the experimental and calculated mole fractions of the second phase.

The binary interaction parameters of NRTL and UNIQUAC models were obtained using ASPEN PLUS V7.2. The van der Waals volume and area parameters are listed in the Table 5. The estimated NRTL and UNIQUAC binary interaction parameters for the systems studied here are presented in Tables 6 and 7, and the calculated tie lines are shown in Figures 1 to 4. α_{ij} (water-solvent) was fixed to 0.2, and the rest were fixed at 0.3. As can be inferred from the root-mean-square error values given in Table 8, a fairly good correlation of the experimental values with NRTL and UNIQUAC was obtained.

The new (liquid–liquid) equilibrium data system was tested with correlations of Hand¹⁵ and Othmer–Tobias,¹⁶ eqs 8 and 9, respectively:

$$\log \left(\frac{x_2^{\text{solvent}}}{x_3^{\text{solvent}}} \right) = a_H \log \left(\frac{x_2^{\text{water}}}{x_1^{\text{water}}} \right) + b_H \quad (8)$$

$$\log\left(\frac{1 - x_3^{\text{solvent}}}{x_3^{\text{solvent}}}\right) = a_{\text{OT}} \log\left(\frac{1 - x_1^{\text{water}}}{x_1^{\text{water}}}\right) + b_{\text{OT}} \quad (9)$$

Symbols a and b are the angular and linear coefficients, and subscripts H and OT refer to Hand and Othmer–Tobias, respectively. Table 9 shows the coefficients of eqs 8 and 9 obtained by linear regression on data presented in Table 3. R^2 values are between 0.912 and 0.987 were achieved for all systems at 298.15 K.

4. CONCLUSIONS

In this work, the experimental determination of the liquid–liquid equilibrium for the ternary systems {water + acrylic acid + (MIBK, toluene, *o*-xylene, *p*-xylene)} at 298.15 K were carried out. The NRTL and UNIQUAC models accurately correlate the experimental data. These results facilitate the LLE data management, and in addition, they might be useful for future simulation of thermodynamical data concerning systems involved in the purification of acrylic acid. The behavior of 4-methyl-2-pentanone, toluene, *o*-xylene, and *p*-xylene on the purification of acrylic acid from its mixtures with water can be compared by means of solute distribution ratios and selectivities. Solute distribution ratios of 4-methyl-2-pentanone are much higher, leading to a more effective removal of the acrylic acid. Selectivities are in three cases higher than one, but values of this parameter reached with aromatics are clearly lower than those found with 4-methyl-2-pentanone.

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Notes

The authors declare no competing financial interest.

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